PRANITH KUMAR DUMPA

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PROFESSIONAL SUMMARY

B.Tech Electronics and Communication Engineering student at Anurag University with a strong foundation in C, Python, Java fundamentals, embedded programming, basic algorithms, problem-solving, debugging, and testing. Hands-on experience developing an Arduino-based autonomous obstacle-avoiding robot using an HC-SR04 ultrasonic sensor, motor driver, and DC motors. Interested in software, embedded systems, and technology internships where I can apply programming fundamentals, learn from experienced teams, and contribute to practical engineering projects.

TECHNICAL SKILLS

Programming Languages: C, Embedded C/C++, Python, Java (Beginner)
Embedded Systems: Arduino, Arduino IDE, HC-SR04 Ultrasonic Sensor, Motor Driver, DC Motors
Computer Science: Arrays, Strings, Basic Algorithms, Logical Problem-Solving
Software Engineering: Clean Coding, Debugging, Testing, Requirements Understanding
Development Tools: Visual Studio Code, MATLAB, Command Line

EDUCATION

Bachelor of Technology (B.Tech) - Electronics and Communication Engineering
Anurag University, Hyderabad
CGPA: 8.5 | Expected Graduation: 2028

PROJECTS

Arduino-Based Obstacle Avoiding Robot Car | Arduino, Embedded C/C++, HC-SR04, Motor Driver, DC Motors

- Developed an autonomous obstacle-avoiding robot car using Arduino, an HC-SR04 ultrasonic sensor, motor driver, and DC motors.
- Programmed the Arduino using Embedded C/C++ to continuously process distance measurements received from the ultrasonic sensor.
- Implemented distance-based control logic to compare sensor readings against a predefined threshold and initiate obstacle-avoidance actions.
- Integrated the HC-SR04 sensor with the Arduino to provide real-time obstacle detection while the robot was moving.
- Controlled DC motor operation through the motor driver based on sensor input, enabling the robot to change direction when an obstacle was detected.
- Tested the robot under different obstacle configurations and refined the control logic to improve autonomous navigation and collision avoidance.
- **Technologies:** Arduino, Embedded C/C++, HC-SR04 Ultrasonic Sensor, Motor Driver, DC Motors, Robotics.

RELEVANT COURSEWORK

Introduction to Computer Science, Programming Fundamentals, Arrays, Strings, Basic Algorithms, Logical Problem-Solving, Software Engineering Fundamentals

TRAINING

Introduction to Computer Science & Programming Fundamentals - Anurag University

- Completed foundational coursework covering programming concepts, computational thinking, and problem-solving.

ACHIEVEMENTS & ACTIVITIES

- Successfully coordinated and executed a college-level event by collaborating with peers and faculty, ensuring smooth planning and on-time completion.
- Maintained a CGPA of 8.5 in the B.Tech Electronics and Communication Engineering program.

CAREER INTERESTS

Software Development Internships | Embedded Systems Internships | Engineering Internships | Python/C Programming Internships | Technology Internships